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EXECUTIVE SUMMARY OF THE THESIS

An Analysis of Discrete Image Tokenizers under Perturbations and Distribution Shift

LAUREA MAGISTRALE IN MATHEMATICAL ENGINEERING - INGEGNERIA MATEMATICA

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1. Introduction

Autoregressive image generation relies on a discrete tokenizer to convert an image into a sequence of codebook indices. The transformer sees only this sequence, so information lost or changed by the tokenizer constrains the generative model. Tokenizers are usually compared by reconstruction quality, yet similar reconstruction scores can hide different representations. A small image perturbation may change many codes, and a local edit may affect distant positions. These properties matter to a model that learns from the token sequence.

This thesis studies the robustness properties of discrete image tokenizers. It focuses on VQGAN [1] and LlamaGen [2], comparing their reconstruction quality, code stability, spatial response, and behavior under distribution shift. Both turn a 256×256 image into a 16×16 token grid with a nominal vocabulary of 16,384 codes. Their common interface holds sequence length and nominal vocabulary size fixed; empirical code use can still differ.

The five research questions examine clean reconstruction and code use (RQ1), global noise (RQ2), token changes outside a perturbed image patch (RQ3), pixel changes outside a directly

edited token patch (RQ4), and reconstruction and code use across datasets (RQ5).

2. Experimental Design

We evaluate the released VQGAN and LlamaGen checkpoints, running inference with controlled perturbations rather than retraining either tokenizer.

The in-distribution reference is the 50,000-image ImageNet validation set [3]. Distribution-shift experiments add ImageNet-V2 [4], ImageNet-Sketch [5], ObjectNet [6], BloodMNIST and OrganAMNIST from MedMNIST v2 [7], and RVL-CDIP document images [8]. These sets span resampling shift, drawing style, viewpoint and background bias, biomedical imagery, and scanned documents.

We report PSNR, SSIM, and LPIPS for paired reconstructions, and FID for distributional similarity within each dataset. For code use, we report the number of active entries, perplexity, and frequency concentration. The following sections introduce each intervention with its results.

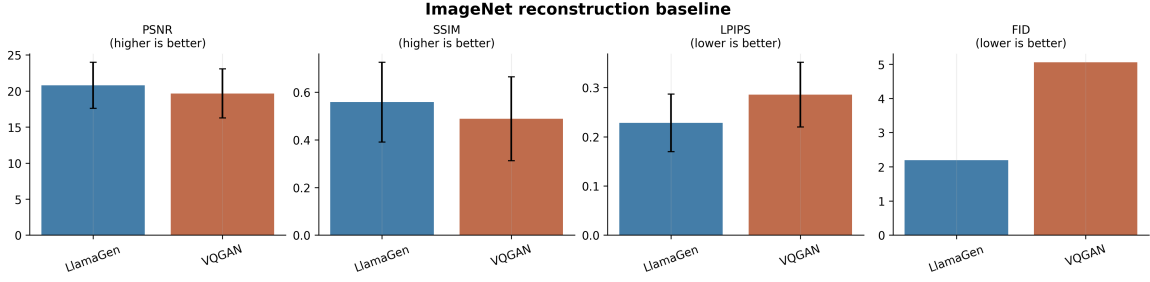


Figure 1: PSNR, SSIM, LPIPS, and FID on the ImageNet validation set. Error bars show the standard deviation of per-image metrics; FID is a dataset-level quantity.

3. Reconstruction and Code Use (RQ1)

Under the recorded exporter settings, LlamaGen has better ImageNet reconstruction scores on the four metrics in Figure 1: PSNR is 20.79 rather than 19.65 dB, SSIM is 0.558 rather than 0.489, LPIPS is 0.228 rather than 0.286, and FID is 2.19 rather than 5.06. A paired 50,000-image PSNR check gives a mean difference of 1.144 dB with a 95% bootstrap interval of [1.140, 1.148] dB; this excludes uncertainty from pre-processing and checkpoint choice.

The two tokenizers reach this result with very different codebook usage. LlamaGen activates all 16,384 entries on ImageNet and has perplexity 16,179. VQGAN activates 972 entries, with perplexity 910. Thus, about 94% of the nominal VQGAN vocabulary is inactive for this checkpoint and dataset. Its top 500 entries account for 66.5% of all assignments, compared with 4.6% for LlamaGen. VQGAN concentrates repeated use in a much smaller vocabulary, while LlamaGen distributes assignments across almost the entire codebook.

Equal nominal codebook size therefore does not imply equally broad empirical use. The released checkpoints expose the same index range, but their active vocabularies differ by more than an order of magnitude. Reconstruction fidelity and codebook use must be reported together.

4. Global Robustness (RQ2)

Global Gaussian noise reveals frequent changes in discrete assignments. Noise standard deviations are specified in the internal $[-1, 1]$ image range: $\sigma = 0.1$ (equivalent to 0.05 in $[0, 1]$ before clipping) flips 79.5% of LlamaGen tokens and 75.1% of VQGAN tokens. At $\sigma = 0.2$,

the rates rise to 91.0% and 89.0%; at $\sigma = 0.5$, both exceed 97%. Because the two quantizers use different latent geometries, these rates describe checkpoint behavior rather than a calibrated cross-model robustness ranking.

At $\sigma = 0.5$, PSNR of the noisy-input reconstruction against the *original clean image* is 16.22 dB for LlamaGen and 14.90 dB for VQGAN. In a separate fixed-seed sample of 2,000 saved image pairs, direct decoded RMSE is lower for LlamaGen at each noise level despite its slightly higher token-flip fraction.

Noise also concentrates code use. Perplexity falls from 16,179 to 6,434 for LlamaGen and from 910 to 560 for VQGAN between the clean condition and $\sigma = 0.5$. Top-100 mass, the fraction of assignments to the 100 most frequent codes, rises from 0.012 to 0.115 and from 0.165 to 0.388, respectively.

The flip rates in Figure 2 and sampled decoded RMSE order the checkpoints differently, so code changes cannot be treated as a calibrated image distance.

5. Encoder Locality (RQ3)

The encoder experiment applies noise to a 128×128 image patch corresponding to an 8×8 token block. A strictly block-local mapping would leave assignments outside that block unchanged. Figure 3 shows outside-patch flip rates of 29.9%, 44.1%, and 52.1% for VQGAN, and 30.2%, 42.8%, and 50.5% for LlamaGen, at $\sigma = 0.1$, 0.25, and 0.5. These local-noise levels differ from the global sweep. Masking the patch instead gives outside rates of 56.3% and 49.5%.

Far field denotes tokens at Manhattan distance at least six cells from the block. At $\sigma = 0.1$, pooled far-field flip rates are 17.4% for VQGAN and 17.8% for LlamaGen.

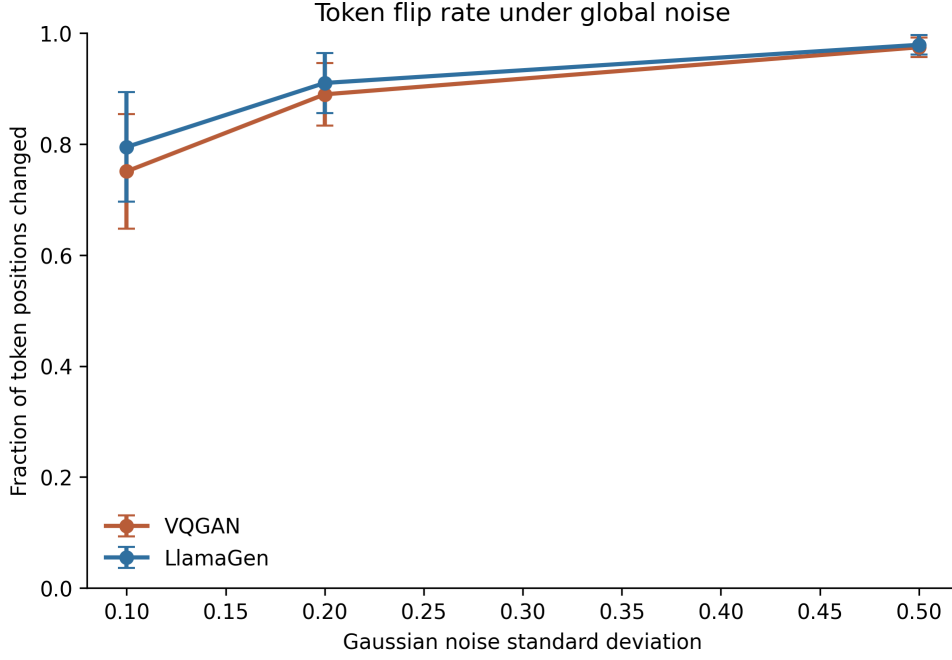


Figure 2: Mean token-flip rate under global Gaussian noise, with σ specified in the $[-1, 1]$ image range. Error bars show one standard deviation across images.

The response extends beyond a one-cell boundary band. Convolutions, attention, normalization, and quantization can contribute, but their roles are not isolated. The measurement establishes dependence beyond the corresponding token block, without claiming changes outside the encoder’s actual receptive field.

6. Decoder Locality (RQ4)

The reverse intervention edits a token patch. Closest and farthest replacements minimize and maximize embedding distance; orthogonal replacement minimizes absolute cosine similarity. For VQGAN, these geometric rules use its ImageNet-active subset; random replacement draws from the full codebook. LlamaGen uses the full codebook for all modes. The following fidelity metrics compare edited and clean reconstructions within the edited patch.

At a 25% edit fraction, closest replacement gives LlamaGen PSNR 20.36 dB and LPIPS 0.134, versus 20.00 dB and 0.182 for VQGAN. Farthest replacement at the same fraction gives 7.74 dB and 0.702 versus 3.74 dB and 0.790 (Figure 4). These edits are not severity-matched across codebooks, so the ordering does not isolate decoder sensitivity.

Decoder response extends outside the edited re-

gion. At 25%, the mean outside response, measured as per-pixel RGB RMSE on $[0, 1]$, ranges from 0.013 to 0.050 for LlamaGen and 0.019 to 0.090 for VQGAN across edit modes. Closest replacement gives the smallest absolute outside response. A 500-image saved-output profile shows response declining with distance but still measurable 80–96 pixels from the boundary.

Repeating the protocol on ImageNet-V2 produces similar aggregate trends. Across modes and patch fractions, the largest LPIPS change is 0.0068 and the largest absolute outside-response change is 0.0016. At the 25% fraction, patch PSNR differs by at most 0.17 dB. Encoder and decoder probes use different interventions and response metrics; both show effects beyond the edited region.

7. Distribution Shift (RQ5)

The final study asks whether a tokenizer uses the same discrete vocabulary when the image distribution changes. LlamaGen retains all ImageNet-active codes on ImageNet-V2 and 16,383 of 16,384 on ImageNet-Sketch, 98.4% on BloodMNIST, and 99.4% on RVL-CDIP. VQGAN retains all 972 entries on ImageNet-V2 and ImageNet-Sketch, 99.4% on BloodMNIST, and 99.9% on RVL-CDIP. A code counts as active

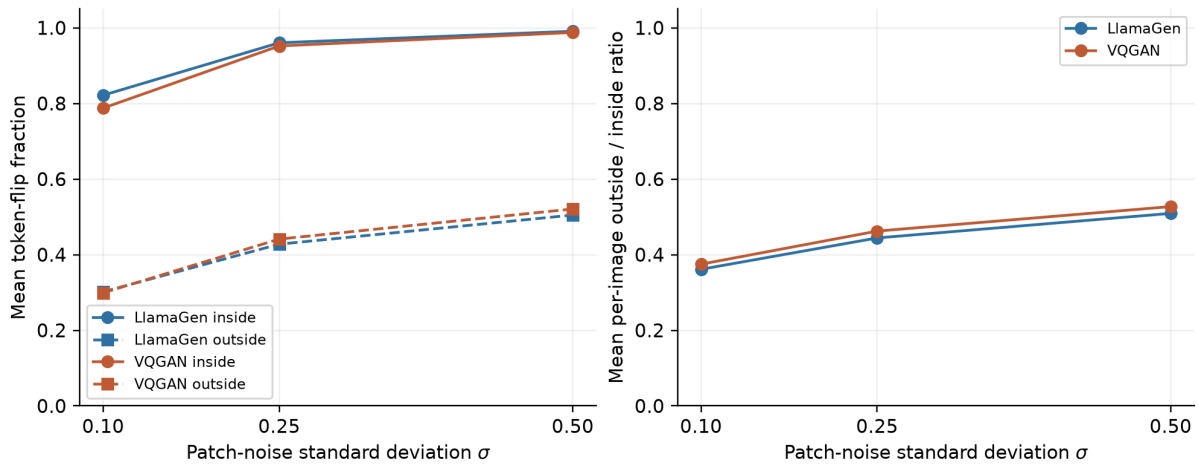


Figure 3: Inside- and outside-patch token responses to local image-space interventions. A substantial outside response remains for both tokenizers.

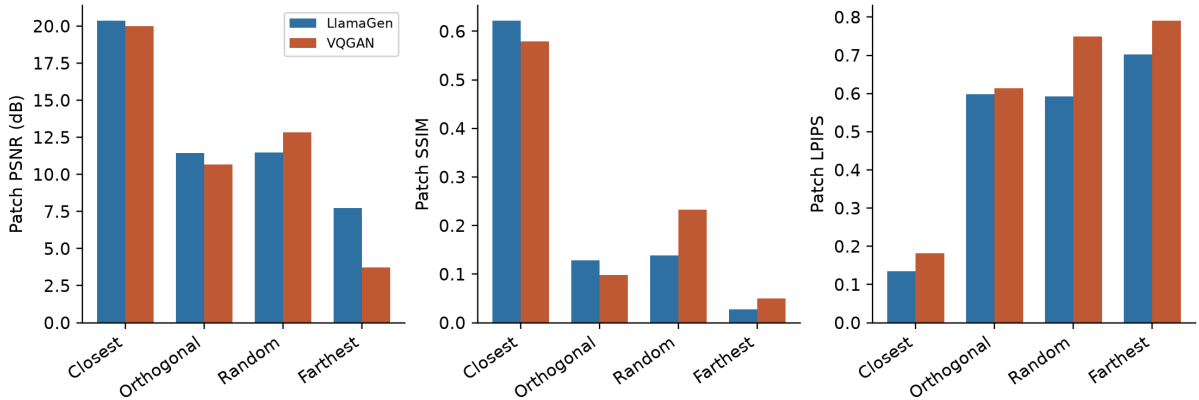


Figure 4: Fidelity within the edited patch relative to the clean reconstruction, at a 25% edit fraction.

after a single assignment, so this overlap alone does not establish stability.

Figure 5 reports Jensen–Shannon divergence (JSD, in nats) and the perplexity ratio to each model’s ImageNet baseline. For LlamaGen, JSD is 0.0020 on ImageNet-V2, 0.1531 on ImageNet-Sketch, 0.2260 on BloodMNIST, and 0.2510 on RVL-CDIP. Its perplexity ratio falls from 0.995 to 0.230 across those datasets. VQGAN shows the same ordering: JSD ranges from 0.0004 to 0.1949, and the ratio falls from 1.001 to 0.325. JSD is descriptive; no sampling null calibrates small values or comparisons between vocabularies.

Reconstruction scores also depend on image content, so their absolute values across datasets do not measure the effect of distribution shift alone. The redistribution finding is evident in ImageNet-Sketch, BloodMNIST, and RVL-CDIP.

Matching the ImageNet reference sample size largely removes the apparent ImageNet-V2 positional-entropy decrease. BloodMNIST and RVL-CDIP retain substantial decreases after the same check. Unequal-size estimates for the other datasets remain descriptive.

Frequency divergence and concentration expose changes that active-set overlap does not.

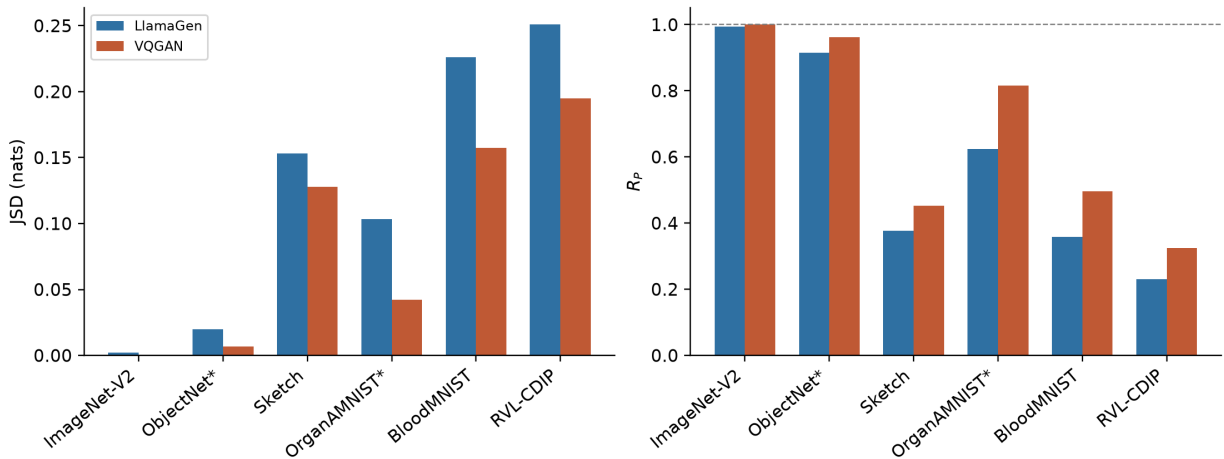


Figure 5: Global code-usage shift relative to ImageNet. JSD is in nats and R_P is the perplexity ratio to each tokenizer’s ImageNet baseline. Asterisks mark the two exports described in Limitations.

8. Limitations

The experiments compare one released checkpoint per tokenizer. Differences in training, codebook geometry, and preprocessing prevent attribution to a single design choice. Noise draws and patch locations were not matched across checkpoints.

ObjectNet retained its red identification border; the local OrganAMNIST export omitted 1,643 official training images. Their results apply to those saved inputs. The study evaluates tokenizers and their reconstructions, without testing downstream autoregressive generation.

9. Conclusions

Across the five research questions, the evidence yields five findings. First, LlamaGen has better ImageNet reconstruction scores under the recorded settings and uses essentially its full codebook, while the evaluated VQGAN checkpoint uses about 972 entries. Second, the tested global noise flips most tokens in both models; fidelity to the original image is a separate measure. Third, local image perturbations change token assignments well outside the perturbed patch. Fourth, local token edits affect pixels outside the corresponding image region. Fifth, ImageNet-Sketch, BloodMNIST, and RVL-CDIP largely preserve the active vocabulary but change its frequency profile. Reconstruction quality, codebook use, token stability, and spatial response answer different questions. The consequences for autoregressive

generation require a separate evaluation. Despite their shared token grid and nominal vocabulary, these checkpoints produce very different code distributions, and both redistribute those codes under shift.

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